

CSCE 638 NATURAL LANGUAGE PROCESSING · MODULE 06 · LECTURE NOTES

The Transformer

Attention, completely.

MODULE OUTCOMES

- Explain attention as a differentiable lookup.
- Explain multi-head attention and what heads do.
- Explain positional encoding and why it is needed.
- Explain the full block and the training objectives.
- Implement the architecture.

1 Attention

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{Q}\mathbf{K}^T / \sqrt{d}) \mathbf{V}$$

- Each query is compared against every key by dot product; the softmax turns those similarities into weights, and the output is a weighted average of the values.
- The $\sqrt{d}$ scaling is not cosmetic. Without it, a large dimension makes the dot products large, which saturates the softmax into a near-one-hot distribution and makes the gradient vanish — so the model cannot learn to attend softly at all.
- And the path length between any two positions is $O(1)$ — which is Module 05 §3's limitation removed outright, and is the quality half of §4's argument.
- Read the whole operation as a lookup: the query asks a question, the keys advertise what each position has to offer, and the values are what actually gets returned — which is the framing that makes Q, K, and V stop looking arbitrary.

ATTENTION IS A DIFFERENTIABLE DICTIONARY LOOKUP, WHICH IS WHY Q, K, AND V ARE THREE DIFFERENT THINGS

In a hard dictionary lookup you have a key and you get exactly one value. Here the query is compared against every key, producing a similarity distribution, and you receive a weighted average of all the values — which is what makes it differentiable and therefore learnable.

So the three projections have genuinely distinct jobs: the key advertises what information a position holds, the query expresses what the current position is looking for, and the value is what is actually transmitted when a match occurs.

Which is precisely why they are three separate matrices — what makes a position worth attending to is not the same thing as what you want from it once you have decided to. A position might advertise 'I am a verb' and transmit something quite different.

And self-attention is simply the case where all three are computed from the same sequence — every position asks every other position a question, which is what makes it a general-purpose sequence operation rather than a mechanism for aligning two sequences, which is what it was originally invented for.

2 Multiple heads

- **One attention distribution can only express one relation** — and a given position may need its syntactic head, its coreferent, and its topical context all at once, which a single softmax cannot provide.
- **So run several attention operations in parallel on lower-dimensional projections, and concatenate the results** — which costs approximately the same as one full-dimensional head, since the dimensions are divided among them.
- **And the observed specialisation is real but loose: some heads reliably attend to the previous token, some to syntactic dependents, some to matching brackets or quotes — and many are not interpretable at all.**
- **With substantial redundancy. A large fraction of heads can be pruned after training with very little loss of performance, which is a serious caution against over-reading any individual head's behaviour.**
- **So treat head interpretations as suggestive rather than explanatory (Module 13 §3). The prunability result is the honest caveat: if most heads can be removed, then the story in which each one performs a specific linguistic function cannot be the whole picture, and the attractive attention-map visualisations should be read with that in mind.**

3 Position

ATTENTION IS PERMUTATION-INVARIANT, SO POSITION MUST BE SUPPLIED EXPLICITLY

The operation treats its input as a *set* — shuffle the positions and the output shuffles identically, with no other change. Which makes 'dog bites man' and 'man bites dog' entirely indistinguishable without additional information, and is a fatal omission rather than a degradation.

So position is injected explicitly: learned embeddings indexed by position, fixed sinusoidal functions, or a relative scheme that modifies the attention scores according to the distance between positions — with rotary embeddings being the current common choice.

And the choice determines how the model handles inputs longer than it was trained on — absolute learned positions do not extrapolate at all, because position 5000 has no embedding if training never reached it, which is why relative and rotary schemes displaced them.

Which is the practical consequence worth remembering: a model's usable context length is a property of its positional scheme at least as much as of its training data — so a claimed context window and a *usable* context window can differ substantially, and the difference is measurable (Module 10).

ONE BLOCK

```
x = x + MultiHeadAttention(LayerNorm(x))
x = x + FeedForward(LayerNorm(x))
```

AND THAT IS ALL. Stack it N times.

WHY EACH PIECE

residual	the gradient path (CSCE 636 M04)
layer norm	stabilises the scale; the pre-norm placement shown here trains more reliably than post-norm does
feedforward	per-position capacity; attention mixes ACROSS positions and this processes WITHIN one. It holds most of the parameters in the model.
stacking	each layer attends over the previous layer's contextual representations, so the representations become progressively more contextual

THE MASK IS WHAT DISTINGUISHES THE VARIANTS

causal mask	-> decoder; each position sees only the past, so it can generate
no mask	-> encoder; fully bidirectional

The mask is the only structural difference between an encoder and a decoder — which is genuinely worth knowing, because the two families are usually described as though they were different architectures when they differ by one boolean. The attention-mixes-across and feedforward-processes-within division is the other thing to retain: it explains why the feedforward

layer holds most of the parameters and why it is the first place people look when trying to locate stored knowledge.

4 Why it won

- **Constant path length**, so long-range dependencies are actually learnable (Module 05 §3) — which is **the quality argument**, and it is real.
- **And full parallelism across positions during training**, which is **the argument that actually mattered: it made training on far more data with far more parameters affordable** at a time when recurrence was the binding constraint on both.
- **Because the scaling result is what produced the capability** (Module 07) — **and the architecture's real contribution was making the necessary scale reachable** rather than being a better model at equal scale, which is a more honest account of its importance.
- **The cost is quadratic attention** in sequence length, in both time and memory — **which is the active research constraint** and the reason for the whole efficient-attention literature (Module 05 §4's note).
- **And a large appetite for data**, since it carries fewer built-in assumptions than a convolution's locality or a recurrence's sequentiality. **Fewer built-in assumptions is a cost at small scale and an advantage at large scale** — which is **the general shape of the inductive bias trade** (CSCE 633 Module 03) and explains why transformers lose to simpler models on small datasets.

AND THE IMPLEMENTATION INSTRUCTION

Implement it once, from the equations, without copying — single-head attention, then multi-head, then one block, then the stack. Four steps, and a long afternoon.

Because the architecture is genuinely simple and reads as complicated — and **the gap closes entirely once you have written the four pieces yourself**, which is the single most worthwhile exercise in this course.

Check your implementation against a reference output, because **a transposed matrix or a misplaced mask produces something that trains badly rather than failing loudly** — the commonest outcome is a model that learns something, just worse, and you will not know why.

Which is CSCE 711 Project 1's test-vector discipline arriving in a different subject — a construction that matches known outputs is correct in a way that one which merely runs is not, and the habit transfers to everything.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Vaswani et al. — Attention Is All You Need (free)	The original. Short, readable, and the ablations in it are more informative than the architecture diagram.
The Illustrated Transformer (free)	§1 through §3 visually — read this first if the equations are not landing.
The Annotated Transformer (free)	§4's implementation instruction, done for you — use it to check your own version rather than to replace writing one.
Michel et al. — Are Sixteen Heads Really Better than One? (free)	§2's prunability result , which is the honest caveat on head interpretation.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Implement single-head attention** from the equation and check it against a reference.
- 2 **Remove the $\sqrt{d}$ scaling** and observe what happens to the softmax and the gradient.
- 3 **Explain the roles of Q, K, and V** in the lookup framing.
- 4 **Implement multi-head attention** and confirm the parameter count matches one full head.
- 5 **Visualise several heads** on a sentence, and find one that is interpretable and one that is not.
- 6 **Prune half the heads** and measure the loss.
- 7 **Shuffle the input positions** with positional encoding removed, and confirm the output is invariant.
- 8 **Compare absolute and relative schemes** on inputs longer than training.
- 9 **Build the full block and stack it**, then train on a small task.
- 10 **Change only the mask** and confirm you have turned an encoder into a decoder.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Write the attention equation and explain each factor.
- 2 Why is the $\sqrt{d}$ scaling necessary?
- 3 Why are Q, K, and V three separate projections?
- 4 Why use multiple heads, and what is actually observed?
- 5 What does the prunability result imply about head interpretation?
- 6 Why is positional information necessary, and what are the three schemes?

- 7 What determines a model's usable context length?
- 8 Give the block, and say why each piece is there.
- 9 What is the only structural difference between an encoder and a decoder?
- 10 Give the two reasons attention won, and say which mattered more.